

Slide 1 – Project Overview

Income Prediction Using Machine Learning

Objective

- Predict whether an individual's annual income exceeds **\$50,000**.
- Support targeted marketing and premium service allocation.
- Reduce unnecessary operational costs by identifying high-income customers accurately.

Business Goal

- Prioritize **Precision** to minimize false positive predictions.

Presenter: Muhammad Usman Ilyas

Slide 2 – Dataset & Data Preparation

Dataset

- Adult Census Income Dataset (OpenML)
- Approximately **48,800 records**
- 14 original features

Key Features

- Age
- Education
- Occupation
- Workclass
- Hours Worked
- Capital Gain/Loss
- Marital Status
- Native Country

Feature Engineering

- Age Bucket
- Hours Bucket
- Capital Gain Flag
- Log Capital Gain

- Higher Education Flag
- Education × Hours Interaction

Slide 3 – Model Development

Models Evaluated

- Logistic Regression
- Random Forest
- HistGradientBoosting
- Stacking Ensemble

Imbalance Handling Compared

- Baseline
- Class Weighting
- Random Oversampling
- Random Undersampling
- SMOTE

Selected Model

 Random Forest

Reason:

- Highest Precision
- Strong ROC-AUC
- Reliable deployment performance

Slide 4 – Final Results

Hold-Out Test Performance

Metric	Result
Accuracy	83.97%
Precision	86.21%
Recall	39.31%
F1 Score	54.00%
ROC-AUC	0.9014

PR-AUC	0.7717
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Decision Threshold

- **0.50**
- Selected because it provides the highest precision while reducing false positives.

Slide 5 – Interpretability & Fairness

Most Important Features

- Education Number
- Capital Gain
- Education
- Occupation
- Hours Worked
- Age
- Native Country

Interpretability

- Feature Importance
- Permutation Importance
- SHAP Explanations

Fairness

Model evaluated across:

- Sex
- Race

Result:

- No major performance disparities observed.
- Continuous monitoring recommended after deployment.

Slide 6 – Deployment & Next Steps

Deployment Artifact

- adult_income_pipeline.pkl
- Production inference script
- Input validation

- Handles unseen categories
- Probability prediction

Monitoring

- Data Drift
- Precision
- Prediction Distribution
- Fairness Metrics
- Missing Values

Next Steps

- Shadow Deployment
- 20% A/B Testing
- Production Rollout
- Retraining every 3 months

Key Takeaways

- Successfully built a production-ready income prediction model.
- Random Forest achieved the best precision (**86.21%**).
- SHAP explanations improve transparency and trust.
- Deployment pipeline and monitoring strategy are ready.
- Recommended phased rollout through A/B testing before full production deployment.

Thank You

Questions?